Economic Complexity HW III. - Kolos Ágoston Mohácsi, Marcell Szarka: ***Inter-firm connections effect on semiconductor patent output, a comparative analysis of East Asia and the United States.***

Problem and gap, based on literature:

Fleming et al. (2007) showed that regional collaboration networks play a key role in innovation productivity - inventor connectivity within the United States matters for patent output. However, the paper didn't analyze whether collaborations happened within or between firms - which is what we plan to focus on.

Based on research conducted on the market dynamics of semiconductor manufacturing and the value chain, Yeung (2022) reports that actors in the industry are heavily dependent on each other and therefore engage in close collaboration, both in the design and manufacturing planning and in R&D, with US firms co-locating in East Asia to take advantage of knowledge spillovers and for a more efficient, competitive process. With this collaboration and inter-firm knowledge sharing and research documented, the question becomes: does this translate to formal shared innovation output in the form of patents?

In the semiconductor industry, formal co-patenting between firms is found to be rare (Molling et al., 2023), only 0.17% of the roughly 1.07 million patents held by the largest 16 semiconductor firms are co-owned. In a similar manner, Wang et al. (2012; 2017) found that patent co-assignee networks barely overlap with patent citation networks, but knowledge spreads better through patent citation (weak ties) than in the form of formal patenting (strong ties). Innovation is concentrated in a few firms, and formal collaboration is mostly typical of East Asian firms. Knowledge still spills over through engineers switching jobs (Almeida, 1996) - it is worth observing the co-inventor ties that may cross firm boundaries. With semiconductors being a salient industry in recent years, policy implications also matter - Rasiah & Wong (2021) demonstrate that advancements in the space are heavily dependent on state support and local university research, which may serve as additional explanation for the differences in US and East Asian patenting, as well as show as a "blueprint" for the United States to follow. A co-inventor network may reveal some differences in underlying knowledge sharing/collaboration that advances our understanding of the industry and its innovation networks.

Our proposal is to create a patent data based collaboration network in the semiconductor sector and measure inventor connections and firm boundaries, expanding on the previous research made in the topic with a network science methodology based approach of empirically evaluating it.

Data & Methodology:

We want to use a database which holds "successful" patent registries with the location, entity, inventors, year of submission and the "field" in which the patent belongs to. Our primary intent is to use the United States Patent and Trademark Office (USPTO, n. d.) "PatentsView Annualized Patent Data" database which satisfies all of these conditions and contains all patents which were granted by the USPTO between 1976 and 2025. While granting patents in the US does not equal to the patent being invented within the US, using the USPTO registry can be biased as it can underrepresent non-US based patents. So we preliminarily compare the USPTO aggregated number of patent grants from East Asia with an international database based aggregated numbers such as WIPO's (n. d). If the results reveal that a significantly large portion of East Asian patents are not present within the USPTO, we will need to switch to another free, but international patent database. In this case we will intend to use WIPO (n. d.) database, which is less standardized and convenient but solves USPTO's bias.

We select regions in East Asia and in the US, with sufficient amount of patent submissions within the field to establish a network. For each region, for each year between 2000-2017 we take 5 year retrograde time frame, constituting a collaboration network for set region and year. Each inventor is a node, collaboration between two inventors is an edge, and edge weights are given by counting the number of collaborations within the timeframe. Each node is given a "characteristic firm" by taking the firm for which the inventor submitted the most patents within the timeframe. In the case firms are tied, the tied firm with the latest patent submission is taken into consideration. From this node characteristics we take the dyadic relation of the nodes, if they match the edge is considered *within-firm*, alternatively it is considered *between-firm*.

Each measure is calculated within the regional network. We replicate Fleming et al.'s 2007 measures: the node proportion of the largest connected component (PCC), the average path length (APL), and the clustering coefficient (CC). Then we make two measures. For checking the presence of *cross-firm* collaborations we calculate the proportion of *cross-firm* edges (PCE). For checking how important are cross-firm collaborations within the network's connectivity, we first remove *cross-firm* edges, then calculate the node proportion of the largest connected component of this restricted network, divide it by PCC, resulting in a Restricted Connectivity

Ratio (RCR). A lower value of RCR suggests higher importance for *cross-firm* collaborations in connectivity. We measure future patent productivity (FPP), by taking the number of patents, divided by the number of inventors, present in the database. We take this measure from a future timeframe of t+1 to t+3 to minimize noise from year-to-year fluctuations. We use future data to avoid the possibility of overlapping calculations.

We estimate a panel regression for FPP. We take both regions and years as fixed effects, and measure the estimated effects of restricted connectivity ratio (RCR) while controlling with the replicated measures of PCC, APL, and CC, the number of inventors (NI), and the proportion of cross-firm innovation (PCE). Controlling with PCE is required to separate the volume of *between-firm* connections by the structural importance of them. To analyze the structural difference between the United States and East Asia, we also create an "East Asia" dummy variable which we interact with the RCR importance for connectivity, thus differentiating the effect of connectivity between East Asian and American innovation networks (RCRI - RCR Interaction). We have to use autocorrelation robust standard errors, as the retrograde timeframe for our measures means that there is overlap within the dependent and explanatory variables thus they are autocorrelated.

Expected results and impact:

Our hypothesis revolve around evaluating the coefficients of RCR (B1) and RCRI (B2). Which goal is to assist policymakers wether the US could replicate the East Asian development model for semiconductors:

- H0: B1 >= 0 (B2 doesn't matter here from the US policy perspective, such the outcomes are residual cases here), implies that the *within-firm* origin of connectivity is not inferior for patent productivity compared to *between-firm* connectivity. This is consistent with the US policies assumption, thus no change is required.
- H1: B1 < 0, but B2 =< 0, implies that *between-firm* connectivity is superior for patent productivity everywhere (at most as strongly as East Asia). This would indicate that imitating the policy of East Asia in actuality isn't sufficient and its success arises from its less formal inter-firm ecosystem. Thus a policy change is required, with the results implying towards implementing further measures which encourage the establishment of *between-firm* ties similarly to East Asia.
- H2: B1 < 0 but B2 > 0, implies that *between-firm* connectivity is superior for patent productivity, but less so in East Asia. This would indicate that the East Asian policy is successful but context specific, thus policy change is warranted but the US should attempt to establish its own version of an inter-firm ecosystem.

Our expectation is H2, as the mechanisms outlined by our literature indicate towards between-firm collaborations mattering, but which (the inferiority of high *within-firm* collaboration) in East Asian nations is compensated with their context specific environmental factors.

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